THE PRESENTATION ON

"Customer Credit Risk & Loan Default Analytics System for Banking Sector"

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Objective

Main Objective:
To develop an intelligent system that predicts customer credit risk and loan default probability using Machine Learning.
Key Objectives:
• **Build a Machine Learning Model**
To analyze customer financial data and predict loan eligibility accurately.
• **Risk Classification of Customers**
To categorize customers into:

- High Risk (likely to default)
- Low Risk (safe customers)

• **Develop Real-Time Prediction Dashboard**
To create an interactive system (using Streamlit) where banks can upload data and get instant results.
• **Enhance Decision-Making for Banks**
To help financial institutions:

- Reduce loan defaults
- Approve reliable customers
- Improve overall lending strategy

Problem Statement

1 Challenges in Banking Sector

•Banks face **high loan default risk**

•Traditional risk assessment methods are:

- ⏳ Time-consuming
- ⚠️ Error-prone

2 Need for Solution

•Requires an **automated & intelligent decision system**

•Faster and more accurate risk evaluation

3 Goal of the Project

•Predict **loan eligibility**

•Analyze and classify **customer credit risk**

Dataset Description

Overview of Dataset

•The dataset contains **loan application records (~1 lakh entries)**

•Used for analyzing customer credit behavior and loan default risk

📄 **Key Features**

•**Credit Score** – Customer's creditworthiness

•**Annual Income** – Yearly earnings

•**Current Loan Amount** – Loan requested

•**Term** – Loan duration

•**Years in Current Job** – Employment stability

•**Home Ownership** – Owned/Rented

•**Purpose** – Reason for loan

•**Monthly Debt** – Existing liabilities

•**Credit History** – Past financial behavior

🎯 **Target Variable**

•**Loan Status**

- 1 → Eligible / Non-default
- 0 → Not Eligible / Default

Requirement Analysis & Planning steps

1: Requirement Analysis
Bank Requirements
•The bank should be able to upload customer loan data into the system
•The system should automatically process the uploaded data
•The system should identify high-risk (risky) and low-risk customers
•The system should generate a report showing customer risk levels

User Requirements (What user wants)
•User should enter details (income, loan amount, credit score)
•System should tell **loan approved or not**
•Easy to use interface (simple form)
•Fast result (within seconds)

System Requirements (What system needs)
•**Software:**

- Python
- Streamlit (for UI)
- Machine Learning Model

2: Planning

Working Steps for home page

1. Bank user insert loan default data
2. Data goes to system
3. ML model processes data
4. System predicts result
5. Result given to the bank

Working Steps (Eligibility check form)

1. User enters his details
2. Data goes to system
3. System apply conditions
4. System send result (decision) to his entered Gmail
5. Result shown to user

Working of report analysis

1. **Data Cleaning –** Handle missing & duplicate values
2. **Data Transformation** – Convert & format data
3. **SQL Analysis** – Query data for insights
4. **Data Visualization** – Create dashboard (Power BI)
5. **Insight Generation** – Identify patterns & trends
6. **Conclusion** – Final findings for decision making

Software Requirement Specification (SRS)

1 Hardware Requirements

•Computer / Laptop
•Minimum 4 GB RAM
•Processor: i3 or above
•Storage: 10 GB free space

2 Software Requirements

•Operating System: Windows / Linux
•Python (for backend & ML model)
•MySQL (for database storage)
•Streamlit (for user interface)
•Power BI (for data visualization & reports)

3.Platforms

Vs code(for UI design)
Jupiter notebook (for ML training)

3: Tools Used

Tools & Technologies

Python → Used for machine learning model and backend logic
MySQL → Used to store customer loan data and summarize data into reports
Streamlit → Used to create a simple and interactive user interface
Power BI → Used to create dashboards and visualize risky customers

System Architecture

Content

1. Home

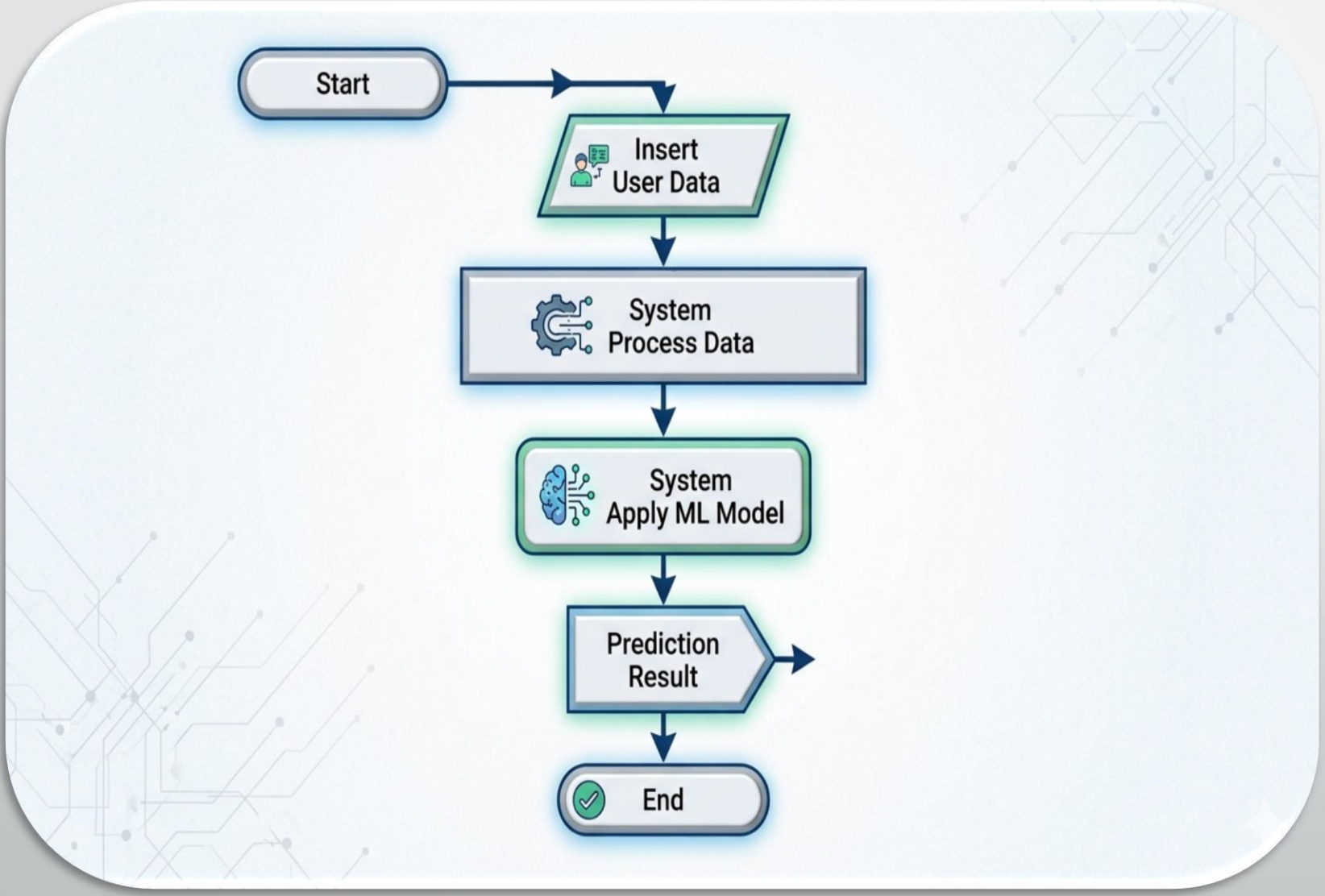
- The system is designed to process bank loan data and predict risky customers
- It follows a simple flow from data input to prediction and report generation
- Machine learning model is used for data prediction

2. Eligibility check form

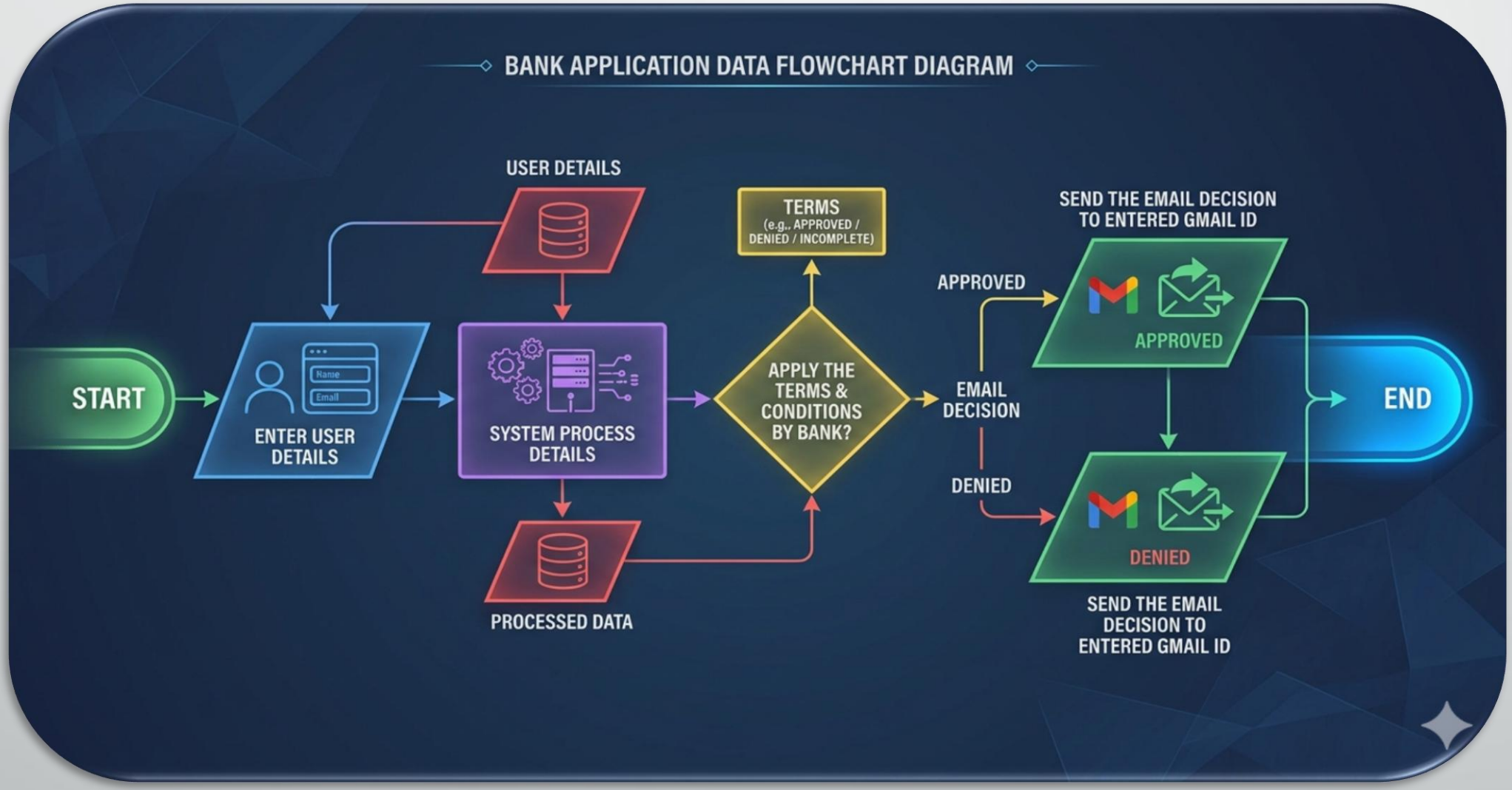
- It helps to know that user is eligible or not to get loan
- It is designed to check the customers eligibility based on banks term and condition
- It helps to know what should be improved to get loan

Working Flow :

Home page:-



Eligibility Check Form



ER Diagram (Database Design):

•This ER diagram represents the database structure of the **Loan Risk Prediction System**
•The system consists of four main entities:

- **Customers** – stores basic customer details like income, job, and home ownership etc.
- **Loans** – contains loan-related information such as loan amount, term, and purpose etc.
- **Credit Info** – stores customer credit history like credit score, accounts, and bankruptcies
- **Financial Info** – includes financial data such as debt and credit balance etc.

Relationships
•Each **customer** is uniquely identified by *customer id*
•A customer can have **loan details**, **credit information**, and **financial records**
•All tables are connected using *customer id* as a common key

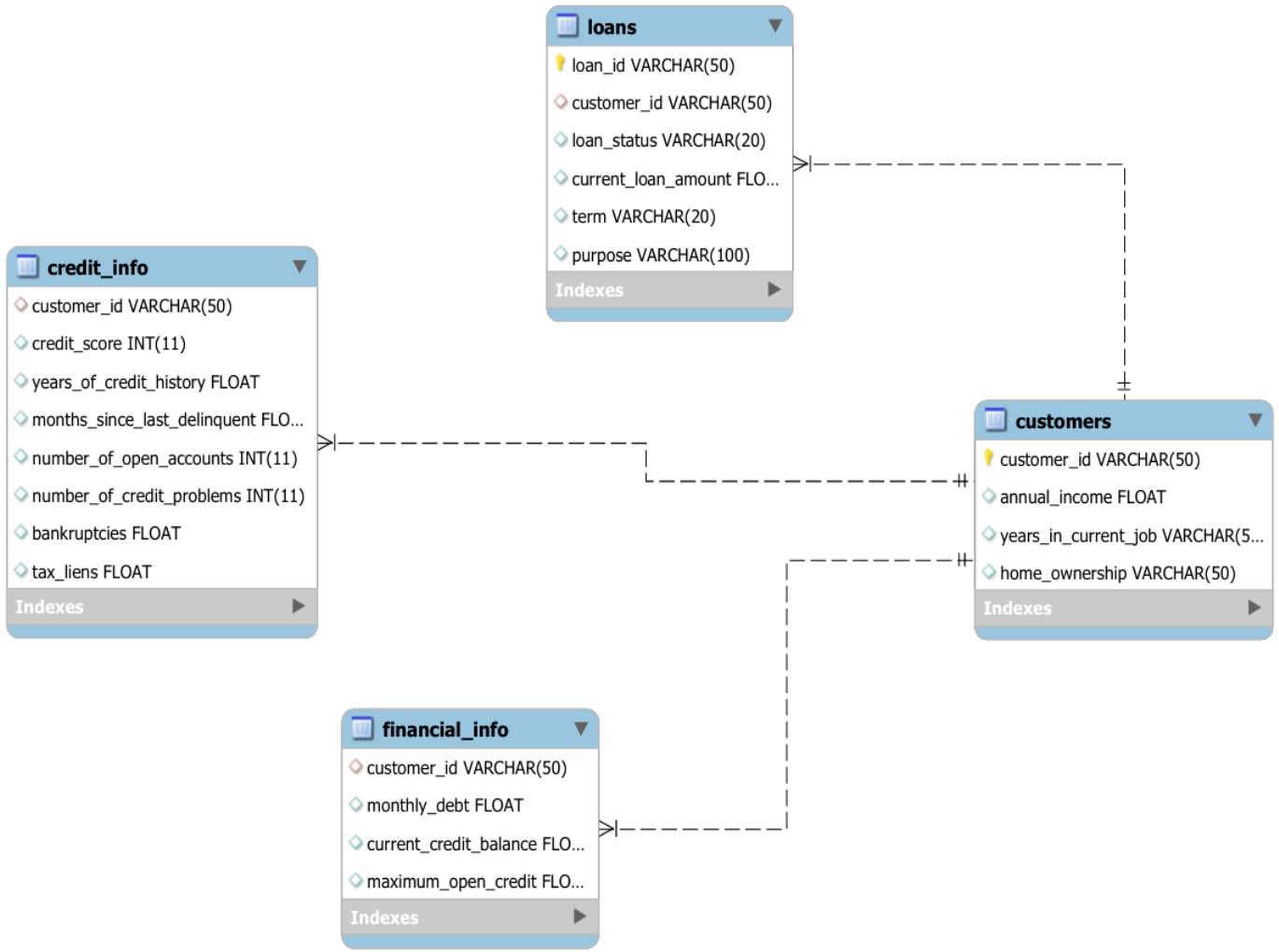
Purpose of Design

- The ER diagram is designed to convert raw loan dataset into a structured database system
- The original dataset contained all customer, loan, and financial details in a single table
- To improve efficiency, the data was **normalized and divided into multiple tables**

How Data is Organized

- **Customers Table** → Stores basic customer information
- **Loans Table** → Stores loan-related details
- **Credit Info Table** → Stores credit history
- **Financial Info Table** → Stores financial data
- All tables are connected using **customer id as a primary key**

ER Diagram (Database Design):



User Interface (UI of project)

•The system provides a simple and user-friendly interface
•Users can input data and view prediction results easily

First step to run UI of customer credit risk and loan default analysis system

•Launch code in vs code editor and run
•In command section you need to write
•{python -m streamlit run app.py}
•Result : Home page

1. Home Page and Prediction result

Working steps:-

1. Take customer loan details data.
2. Insert data using home page browse file
3. Then you can see the data preview of Imported data
4. Then check data and click on Run Prediction
5. After sometime you will get result like eligible or not and high risk and low risk
6. Then click on the prediction result in sidebar
7. You will get prediction report
8. You can download and also can view the dashboard of predicted data

app

Dashboards

Eligibilty check

Navigation

Go to

- Home Dashboard
- Prediction Results

Smart Loan Eligibility Prediction System

Upload Loan Dataset (.csv)

Drag and drop file here
Limit 200MB per file

Browse files

customer_bank_details_data.csv 16.0MB

	Loan ID	Customer ID	Loan Status	Current Loan Amount
8	8af915d9-9e91-44a0-b5a2-564a45c12089	af534dea-d27e-4fd6-9de8-efaa52a78ec0	Fully Paid	548746
9	0b1c4e3d-bd97-45ce-9622-22732fcdc9a0	235c4a43-dadf-483d-aa44-9d6d77ae4583	Fully Paid	215952
10	32c2e48f-1ba8-45e0-a530-9a6622c18d9c	0de7bcdb-ebf4-4608-ba39-05f083f855b6	Fully Paid	0
11	fa096848-6143-4907-b2cf-852a0b06171c	aa0a6a22-a95e-48e0-ba4f-b83456d424e4	Fully Paid	541970
12	403d7235-0284-4bb6-919a-09402fecbf7b	11581f68-de3c-49d8-80d9-22268ebb323b	Fully Paid	0
13	01d878ae-efa2-41e2-8159-6c834ff09f47	900c9191-2c20-4688-af7e-07c59b5d5a24	Fully Paid	0
14	2e841c8f-3dc1-464d-91c1-3d3d51e64c38	2ac05980-7848-4692-89ae-9321afe650f8	Fully Paid	234124
15	7cbaa3fa-16fd-4343-9bcb-e90b34a1072f	3ec886e7-f15d-4c35-83d0-bdec4817ae4b	Fully Paid	449020
16	c9a16a9d-8801-4430-b445-dbf9cf845e31	abb4c446-08ea-49ff-aeb8-5e1e9da673e7	Charged Off	653004
17	24e8c8bd-d10b-4dac-8b81-1da470ff5ecb	967e8733-7189-49b7-a3ab-6a1d0e1abdac	Fully Paid	666204
18	c6be21f0-80b1-46b3-8019-16646fd2137d	c67b2cb5-9f91-4bcb-9a03-03d1589c6c1a	Fully Paid	66396

Run Prediction

Eligible
2410

Not Eligible
87375

High Risk
87375

Low Risk
2410

Prediction Completed

app

Dashboards

Fligibilty check

Navigation

Go to

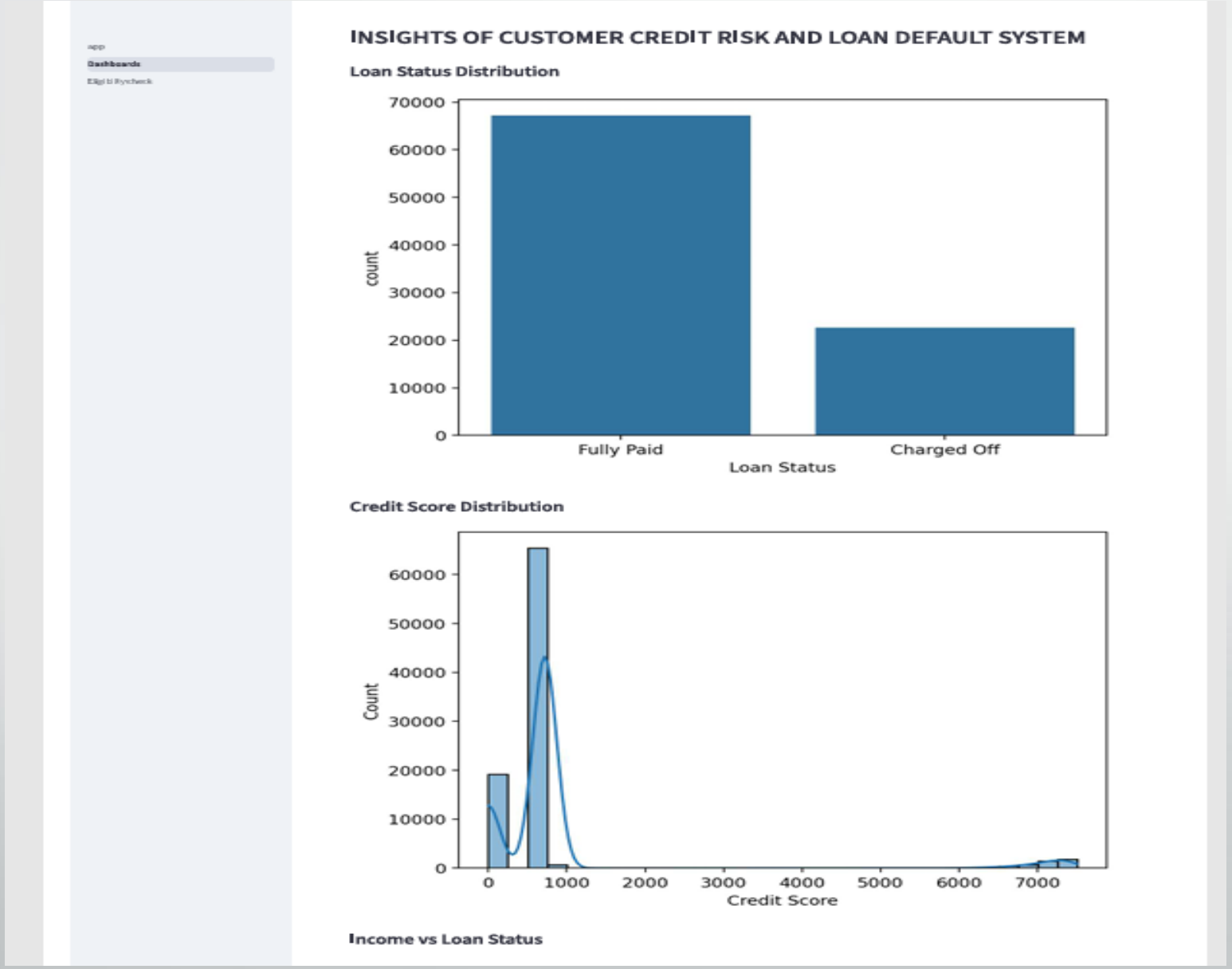
- Home Dashboard
- Prediction Results

Prediction Results

	Loan ID	Customer ID	Loan Status	Current Loan Amount
0	14dd8831-6af5-400b-83ec-68e61888a048	981165ec-3274-42f5-a3b4-d104041a9ca9	Fully Paid	445412
1	4771cc26-131a-45db-b5aa-537ea4ba5342	2de017a3-2e01-49cb-a581-08169e83be29	Fully Paid	262328
2	4eed4e6a-aa2f-4c91-8651-ce984ee8fb26	5efb2b2b-bf11-4dfd-a572-3761a2694725	Fully Paid	0
3	77598f7b-32e7-4e3b-a6e5-06ba0d98fe8a	e777faab-98ae-45af-9a86-7ce5b33b1011	Fully Paid	347666
4	d4062e70-befa-4995-8643-a0de73938182	81536ad9-5ccf-4eb8-befb-47a4d608658e	Fully Paid	176220
5	89d8cb0c-e5c2-4f54-b056-48a645c543dd	4ffe99d3-7f2a-44db-afc1-40943f1f9750	Charged Off	206602
6	273581de-85d8-4332-81a5-19b04ce68666	90a75dde-34d5-419c-90dc-1e58b04b3e35	Fully Paid	217646
7	db0dc6e1-77ee-4826-acca-772f9039e1c7	018973c9-e316-4956-b363-67e134fb0931	Charged Off	648714
8	8af915d9-9e91-44a0-b5a2-564a45c12089	af534dea-d27e-4fd6-9de8-efaa52a78ec0	Fully Paid	548746
9	0b1c4e3d-bd97-45ce-9622-22732fcdc9a0	235c4a43-dadf-483d-aa44-9d6d77ae4583	Fully Paid	215952

Download Report

Dashboard output:



Report analysis:

Steps Performed

- Missing values handled (Income, Credit Score etc.)
- Duplicate records removed
- Data type conversion (string → numeric)
- Irrelevant columns removed (Customer ID, Loan ID)
- Feature formatting (Years in job, Term)

Tools Used

- Excel

Before report data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Loan ID	Customer	Loan Statu	Current Lo	Term	Credit Sco	Annual Inc	Years in cu	Home Owi	Purpose	Monthly D	Years of C	Months sir	Number of	Number of	Current Cr	Maximum	Bankruptci	Tax Liens	Prediction	Probability	Risk
2	14dd8831-	981165ec-	Fully Paid	445412	Short Term	709	1167493	8 years	Home Mo	Home Imp	5214.74	17.2		6	1	228190	416746	1	0	0	36.11	High Risk
3	4771cc26-	2de017a3-	Fully Paid	262328	Short Term			10+ years	Home Mo	Debt Cons	33295.98	21.1	8	35	0	229976	850784	0	0	0	44.78	High Risk
4	4eed4e6a-	5efb2b2b-	Fully Paid	0	Short Term	741	2231892	8 years	Own Hom	Debt Cons	29200.53	14.9	29	18	1	297996	750090	0	0	0	0.23	High Risk
5	77598f7b-	e777faab-	Fully Paid	347666	Long Term	721	806949	3 years	Own Hom	Debt Cons	8741.9	12		9	0	256329	386958	0	0	0	36.16	High Risk
6	d4062e70-	81536ad9-	Fully Paid	176220	Short Term			5 years	Rent	Debt Cons	20639.7	6.1		15	0	253460	427174	0	0	0	56.04	High Risk
7	89d8cb0c-	4ffe99d3-	Charged O	206602	Short Term	7290	896857	10+ years	Home Mo	Debt Cons	16367.74	17.3		6	0	215308	272448	0	0	1	100	Low Risk
8	273581de-	90a75dde-	Fully Paid	217646	Short Term	730	1184194	< 1 year	Home Mo	Debt Cons	10855.08	19.6	10	13	1	122170	272052	1	0	0	30.93	High Risk
9	db0dc6e1-	018973c9-	Charged O	648714	Long Term			< 1 year	Home Mo	Buy House	14806.13	8.2	8	15	0	193306	864204	0	0	0	54.59	High Risk
10	8af915d9-	af534dea-	Fully Paid	548746	Short Term	678	2559110	2 years	Rent	Debt Cons	18660.28	22.6	33	4	0	437171	555038	0	0	0	48.98	High Risk
11	0b1c4e3d-	235c4a43-	Fully Paid	215952	Short Term	739	1454735	< 1 year	Rent	Debt Cons	39277.75	13.9		20	0	669560	1021460	0	0	0	42.13	High Risk
12	32c2e48f-	0de7bcdb-	Fully Paid	0	Short Term	728	714628	3 years	Rent	Debt Cons	11851.06	16	76	16	0	203965	289784	0	0	0	11.74	High Risk
13	fa096848-	aa0a6a22-	Fully Paid	541970	Short Term			10+ years	Home Mo	Home Imp	23568.55	23.2		23	0	60705	1634468	0	0	1	59.91	Low Risk
14	403d7235-	11581f68-	Fully Paid	0	Short Term	740	776188	< 1 year	Own Hom	Debt Cons	11578.22	8.5	25	6	0	134083	220220	0	0	0	1.39	High Risk
15	01d878ae-	900c9191-	Fully Paid	0	Short Term	743	1560907	4 years	Rent	Debt Cons	17560.37	13.3		10	1	225549	496474	1	0	0	0.4	High Risk
16	2e841c8f-	2ac05980-	Fully Paid	234124	Short Term	727	693234	10+ years	Rent	Debt Cons	14211.24	24.7	46	10	1	28291	107052	1	0	1	75.23	Low Risk
17	7cbaa3fa-	3ec886e7-	Fully Paid	449020	Long Term			9 years	Own Hom	Debt Cons	18904.81	19.4		8	0	334533	428956	0	0	1	88.79	Low Risk
18	c9a16a9d-	abb4c446-	Charged O	653004	Long Term			7 years	Home Mo	Debt Cons	14537.09	20.5		9	0	302309	413754	0	0	1	78.54	Low Risk
19	24e8c8bd-	967e8733-	Fully Paid	666204	Long Term	723	1821967	10+ years	Home Mo	Debt Cons	17612.24	22	34	15	0	813694	2004618	0	0	0	21.51	High Risk
20	c6be21f0-	c67b2cb5-	Fully Paid	66396	Short Term			10+ years	Rent	Debt Cons	9898.81	27.1		23	1	9728	402380	1	0	0	10.54	High Risk
21	41f7dd8d-	422f9b72-	Fully Paid	390390	Short Term	747	1791738	8 years	Home Mo	Home Imp	2478.55	22.7		6	0	121182	801812	0	0	0	6.52	High Risk
22	150ebbad-	40f729c9-	Charged O	317108	Long Term	687	1133274	8 years	Rent	Debt Cons	9632.81	17.4	53	4	0	60287	126940	0	0	1	67.42	Low Risk
23	31ae42f6-	016c5139-	Fully Paid	128238	Short Term	750	1354073	< 1 year	Rent	Debt Cons	13202.15	11.9		7	0	131936	458788	0	0	0	26.06	High Risk
24	c7e2b784-	5b53e176-	Charged O	153252	Short Term	714	1890690	2 years	Rent	Debt Cons	21900.35	15.7		12	0	891594	1081014	0	0	0	33.3	High Risk
25	14c61ec2-	eb166545-	Fully Paid	91894	Short Term	724	850383	10+ years	Home Mo	Debt Cons	5860.74	17.5		7	0	95608	230626	0	0	0	43.85	High Risk
26	7aa649ff-	54f57722-	Fully Paid	244926	Long Term	704	1249953	4 years	Home Mo	Debt Cons	6812.26	14.4		6	1	143051	245014	0	1	0	37.29	High Risk
27	578c8e10-	cbe53e22-	Fully Paid	465410	Long Term	688	1722654	3 years	Rent	Buy House	15647.45	22.3	30	7	0	107559	488356	0	0	0	50.53	High Risk

loan_prediction_report (8)

After cleaning report

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Customer	Loan Statu	Current Lc	Term	Credit Sco	Annual Income	Home Ow	Purpose	Monthly D	Years of C	Months si	Number o	Number o	Current Ci	Maximum	Bankruptc	Tax Liens	Prediction	Probabilit	Risk
2	981165ec-	Fully Paid	445412	Short Terr	709	1167493	Mortgage	Home Imp	5214.74	17.2	0	6	1	228190	416746	1	0	0	36.11	High Risk
3	2de017a3	Fully Paid	262328	Short Terr	0	0	Mortgage	Debt Cons	33295.98	21.1	8	35	0	229976	850784	0	0	0	44.78	High Risk
4	5efb2b2b-	Fully Paid	0	Short Terr	741	2231892	Own Hom	Debt Cons	29200.53	14.9	29	18	1	297996	750090	0	0	0	0.23	High Risk
5	e777faab-	Fully Paid	347666	Long Term	721	806949	Own Hom	Debt Cons	8741.9	12	0	9	0	256329	386958	0	0	0	36.16	High Risk
6	81536ad9	Fully Paid	176220	Short Terr	0	0	Rent	Debt Cons	20639.7	6.1	0	15	0	253460	427174	0	0	0	56.04	High Risk
7	4ffe99d3-7	Charged C	206602	Short Terr	7290	896857	Mortgage	Debt Cons	16367.74	17.3	0	6	0	215308	272448	0	0	0	100	Low Risk
8	90a75dde	Fully Paid	217646	Short Terr	730	1184194	Mortgage	Debt Cons	10855.08	19.6	10	13	1	122170	272052	1	0	0	30.93	High Risk
9	018973c9-	Charged C	648714	Long Term	0	0	Mortgage	Buy House	14806.13	8.2	8	15	0	193306	864204	0	0	0	54.59	High Risk
10	af534dea-	Fully Paid	548746	Short Terr	678	2559110	Rent	Debt Cons	18660.28	22.6	33	4	0	437171	555038	0	0	0	48.98	High Risk
11	235c4a43-	Fully Paid	215952	Short Terr	739	1454735	Rent	Debt Cons	39277.75	13.9	0	20	0	669560	1021460	0	0	0	42.13	High Risk
12	0de7bcdb	Fully Paid	0	Short Terr	728	714628	Rent	Debt Cons	11851.06	16	76	16	0	203965	289784	0	0	0	11.74	High Risk
13	aa0a6a22	Fully Paid	541970	Short Terr	0	0	Mortgage	Home Imp	23568.55	23.2	0	23	0	60705	1634468	0	0	0	59.91	Low Risk
14	11581f68-	Fully Paid	0	Short Terr	740	776188	Own Hom	Debt Cons	11578.22	8.5	25	6	0	134083	220220	0	0	0	1.39	High Risk
15	900c9191-	Fully Paid	0	Short Terr	743	1560907	Rent	Debt Cons	17560.37	13.3	0	10	1	225549	496474	1	0	0	0.4	High Risk
16	2ac05980-	Fully Paid	234124	Short Terr	727	693234	Rent	Debt Cons	14211.24	24.7	46	10	1	28291	107052	1	0	0	75.23	Low Risk
17	3ec886e7-	Fully Paid	449020	Long Term	0	0	Own Hom	Debt Cons	18904.81	19.4	0	8	0	334533	428956	0	0	0	88.79	Low Risk
18	abb4c446	Charged C	653004	Long Term	0	0	Mortgage	Debt Cons	14537.09	20.5	0	9	0	302309	413754	0	0	1	78.54	Low Risk
19	967e8733-	Fully Paid	666204	Long Term	723	1821967	Mortgage	Debt Cons	17612.24	22	34	15	0	813694	2004618	0	0	0	21.51	High Risk
20	c67b2cb5-	Fully Paid	66396	Short Terr	0	0	Rent	Debt Cons	9898.81	27.1	0	23	1	9728	402380	1	0	0	10.54	High Risk
21	422f9b72-	Fully Paid	390390	Short Terr	747	1791738	Mortgage	Home Imp	2478.55	22.7	0	6	0	121182	801812	0	0	0	6.52	High Risk
22	40f729c9-5	Charged C	317108	Long Term	687	1133274	Rent	Debt Cons	9632.81	17.4	53	4	0	60287	126940	0	0	0	67.42	Low Risk
23	016c5139-	Fully Paid	128238	Short Terr	750	1354073	Rent	Debt Cons	13202.15	11.9	0	7	0	131936	458788	0	0	0	26.06	High Risk
24	5b53e176	Charged C	153252	Short Terr	714	1890690	Rent	Debt Cons	21900.35	15.7	0	12	0	891594	1081014	0	0	0	33.3	High Risk
25	eb166545	Fully Paid	91894	Short Terr	724	850383	Mortgage	Debt Cons	5860.74	17.5	0	7	0	95608	230626	0	0	0	43.85	High Risk

LOAN PREDICTION REPORT DATA

Analysis in SQL:

1 Loan status distribution

- SELECT loan_status, COUNT(*) AS total
- FROM loan_data
- GROUP BY loan_status;

```
--High Risk Customers
SELECT * FROM loan_data
WHERE Credit_Score < 600;

-- Loan Status Count
SELECT Loan_Status, COUNT(*)
FROM loan_data
GROUP BY Loan_Status;

-- Average Income
SELECT AVG(Annual_Income)
FROM loan_data;

-- Average Income by loan status
SELECT    loan_status,    AVG(annual_income) AS avg_income
FROM loan_data
GROUP BY loan_status;

-- Debt vs default
SELECT    loan_status,
AVG(monthly_debt) AS avg_debt
FROM loan_data
GROUP BY loan_status;
```

Customer credit risk and loan default system

Avg Income by Loan Status	
loan_status	avg_income
Charged Off	963764.2105
Fully Paid	1122362.293

count per risky customer	
risk_category	total_customers
High Risk	21139
Low Risk	6508
Medium Risk	62138

credit score impact	
loan_status	avg_credit_score
Charged Off	1826.2083
Fully Paid	573.7969

top fectors cousing default	
purpose	total_defaults
Debt Consolidation	17918
other	2026
Home Improvements	1163
Business Loan	478
Medical Bills	278
Buy a Car	203
Buy House	140
small_business	119
Take a Trip	115
major_purchase	82
moving	43
vacation	29
wedding	23
Educational Expenses	18
renewable_energy	4

debt vs default(Charged off)	
loan_status	avg_debt
Charged Off	18650.56549
Fully Paid	18311.38251

Default vs risky customers		
risk_category	loan_status	total
High Risk	Charged Off	6145
High Risk	Fully Paid	14994
Low Risk	Charged Off	4637
Low Risk	Fully Paid	1871
Medium Risk	Charged Off	11857
Medium Risk	Fully Paid	50281

Avg income by loan status | credit utilization | final risk flag | Credit score risky customers | Loan Approval

Eligibility form:

Working Steps :

1. Fill your all detail asked in the form
2. System will store data and process according banking rules and conditions like:

   credit_score >= 650,
   income >= 30000,
   dti <= 0.4,
   job_years >= 1,
   credit_problems == 0,
   bankruptcies == 0,
   tax_liens == 0,
   credit_history >= 5

3.Then send email result (decision) to Registered Gmail from your detail

app

Dashboards

Eligibilty check

START Eligibility check Form

Personal Details | Income & Job | Credit Details | Result

Your details

Customer Name

Enter full name

Email address

e.g. name@example.com

Annual Income

0

Monthly Debt

0

Credit Problems

0

Tax Liens

0

Credit Score

300

Years in Current Job

0

Bankruptcies

0

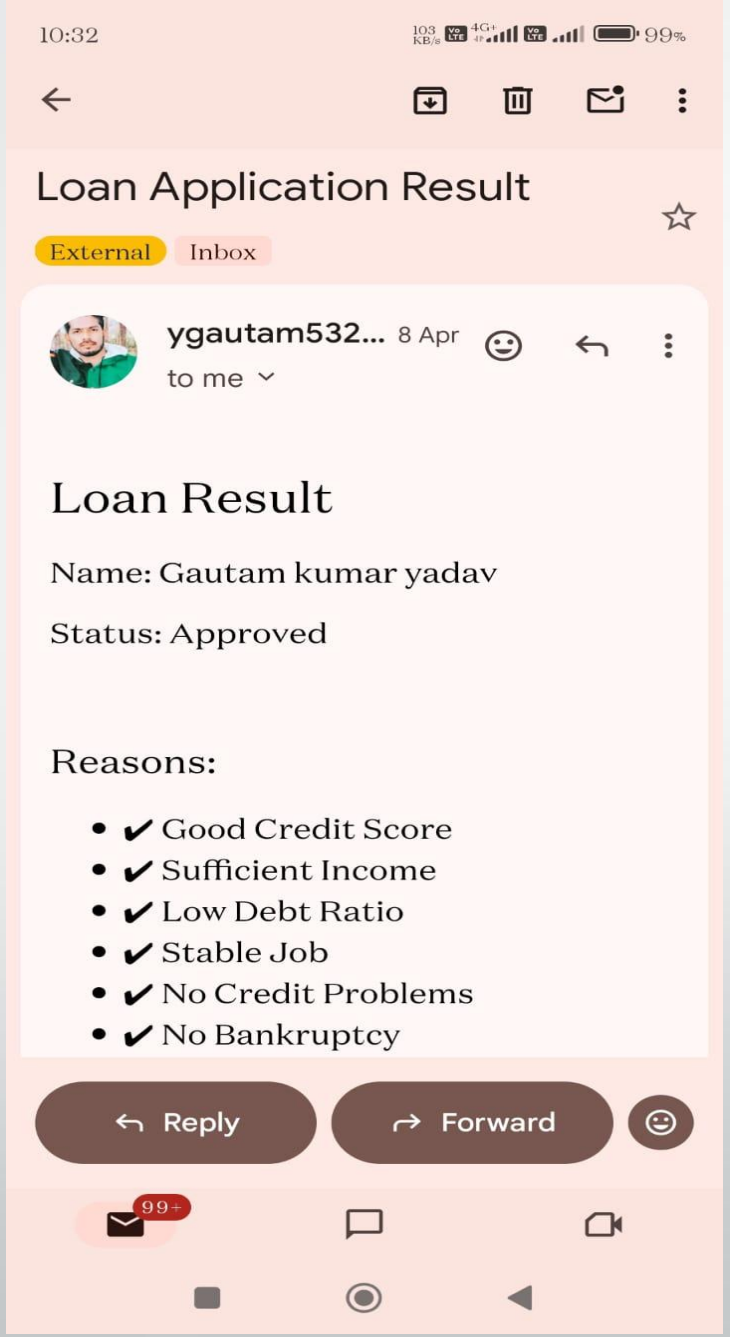
Years of Credit History

0

CONFIRM DETAILS

CHECK ELIGIBILITY

Output:


10:32
99%
Loan Application Result
External
Inbox
ygautam532... 8 Apr
to me
Loan Result
Name: Gautam kumar yadav
Status: Approved
Reasons:
✔ Good Credit Score
✔ Sufficient Income
✔ Low Debt Ratio
✔ Stable Job
✔ No Credit Problems
✔ No Bankruptcy
Reply
Forward
99+

Code Explanation:

1. Core Prediction Logic (ML Model)

The system uses a trained Machine Learning model to predict loan eligibility.

Code:

```
y_prob = model.predict_proba(X.values)[:,1]
threshold = 0.57
predictions = (y_prob > threshold).astype(int)
```

Explanation:

predict_proba() → gives probability of loan approval

Threshold 0.57 used to decide:

0.57 → ✅ Eligible

≤ 0.57 → ❌ Not Eligible

2.Risk classification logic:

Code:

```
def classify_risk(prob):
    if prob > 0.57:
        return "Low Risk"
    else:
        return "High Risk"
```

Output:
•High probability → Low Risk Customer
•Low probability → High Risk Customer

Data Preprocessing (Important Step)

```
data["Credit Score"] = data["Credit Score"].fillna(650)
data = data.fillna(0)
data = data.reindex(columns=model_columns, fill_value=0)
```

Explanation:
•Missing values handled
•Data cleaned & converted into numeric format
•Columns aligned with trained model

Eligibility Form Logic (Rule-Based)

```
if credit_score >= 650 and income >= 30000 and dti <= 0.4:
    result = "Approved"
elif credit_score < 600:
    result = "Rejected"
else:
    result = "Conditionally Approved"
```

Based on:

- Credit Score
- Income
- Debt-to-Income Ratio
- Job Stability
- Credit History

System also sends **email notification**:

```
server.send_message(msg)
```

✓ User receives loan decision on email

Test Cases for Loan Prediction System

Test Case	Input Condition	Expected Output	Actual Output
TC-01	Credit Score = 750, Income = 80000, Debt = Low	Approved	Approved
TC-02	Credit Score = 580, Income = 25000, Debt = High	Rejected	Rejected
TC-03	Credit Score = 680, Income = 40000, Debt = Medium	Conditionally Approved	Conditionally Approved
TC-04	Missing Credit Score	System handles & assigns default value	Handled Correctly
TC-05	Negative Income (-5000)	Invalid Input Error	Error Shown
TC-06	Credit Score = 720, Bankruptcy = 1	Rejected	Rejected